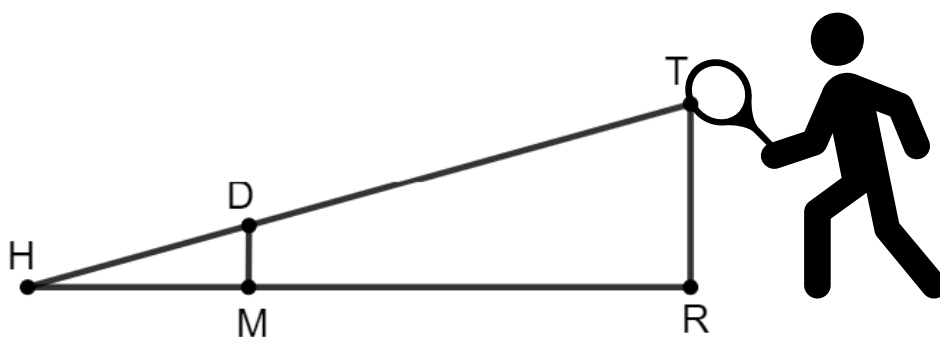


6.13.2 Activity

Problem A

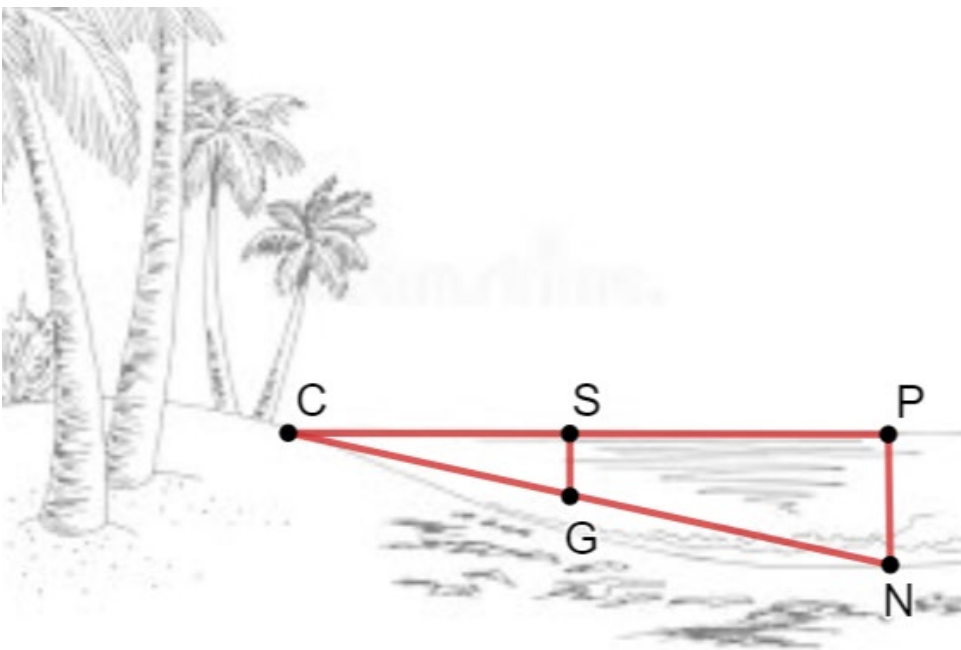
Serena is doing a tennis drill where she wants to hit the ball so that it will just pass over the net, $\overline{DM}$, and lands 6 meters (m) away from the base of the net.



Given that $\triangle HDM \sim \triangle HTR$, $HM = 6$ m, $DM = 0.9$ m, and $MR = 9$ m, what is the height, TR , of the tennis racket?

Problem B

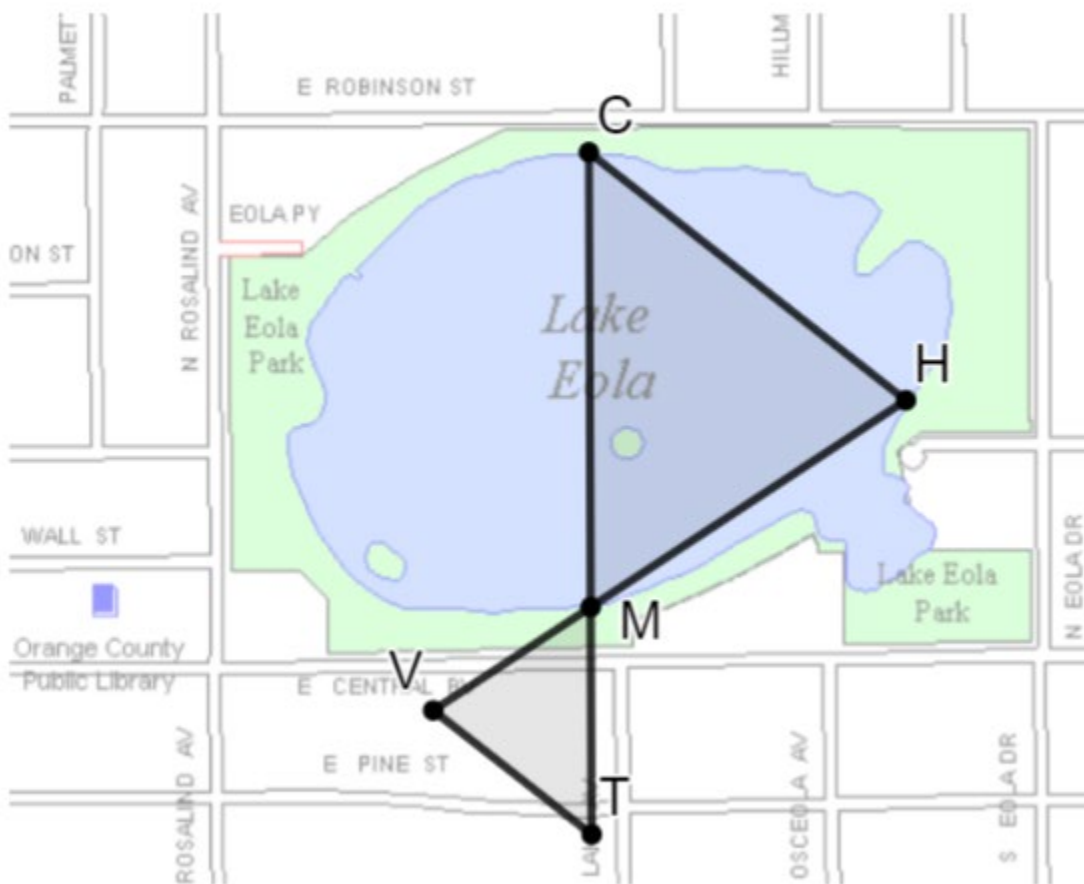
A lifeguard has to place a buoy at the location of the maximum distance and depth that swimmers can swim. $\triangle CPN$ represents the maximum distance and depth swimmers are allowed to reach at the beach, where P is the location of the buoy.



Given that $\triangle CSG \sim \triangle CPN$, $CS = 12$ feet (ft), $SG = (4w - 5)$ ft, $PN = (3w + 3)$ ft, and $CP = 36$ ft, what is the maximum depth, PN , swimmers can swim?

Problem C

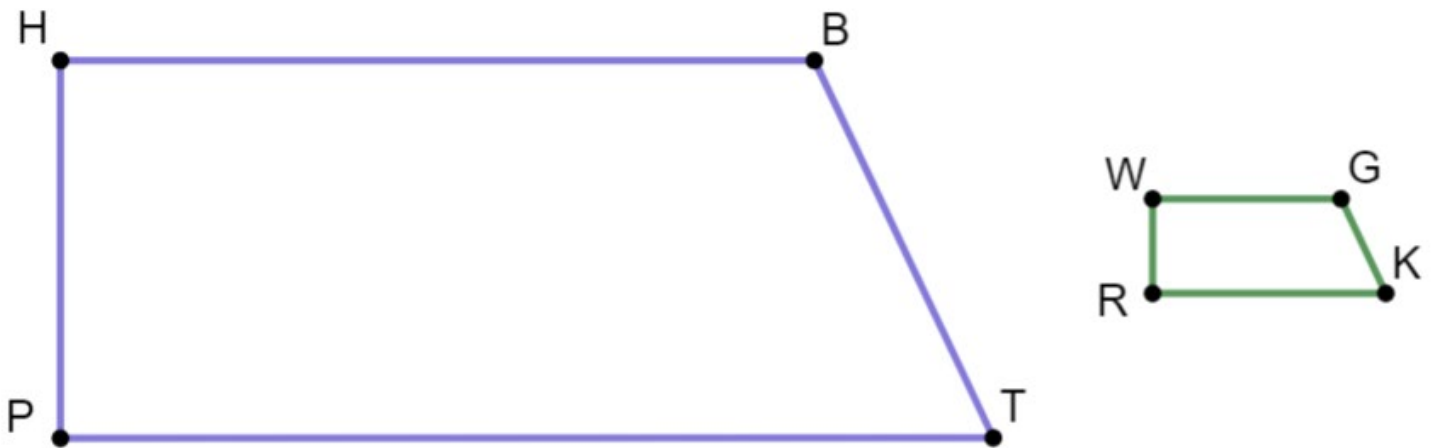
Lake Eola is located in Downtown Orlando. Kelly created two similar triangles such that $\overline{CH} \parallel \overline{VT}$ to help her find the distance across the lake, CM .



Kelly knows that $HM = 855$ feet (ft), $MT = 527.5$ ft, and $VM = 427.5$ ft. What is the distance, in feet, across Lake Eola, CM ?

Problem D

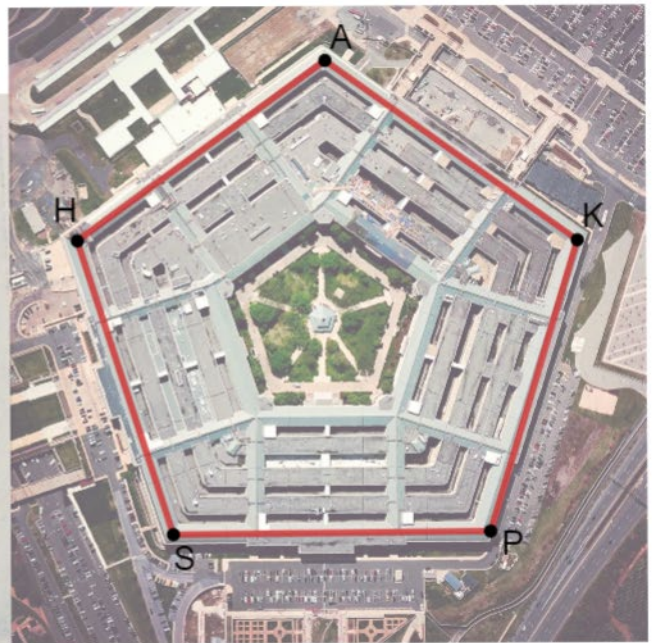
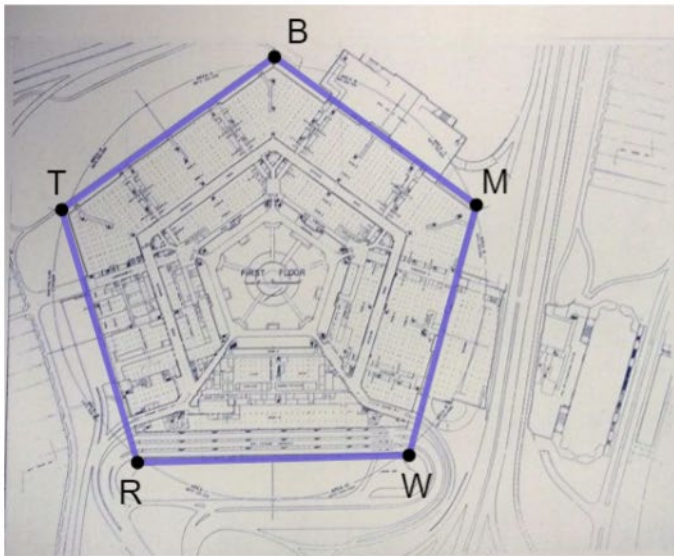
Tyler, a landscape planner, has been hired to create a garden. Tyler's client would like him to create two similar gardens: one garden, $HBTP$, for his backyard, and a similar-sized garden, $WGKR$, for his front yard.



Tyler knows that $PT = 74$ feet (ft), $HP = (3x + 9)$ ft, $WG = 15$ ft, $WR = (2x - 6.5)$ ft, $BT = 32$ ft, and $RK = 18.5$ ft. What is the perimeter of $HBTP$, the backyard garden?

Problem E

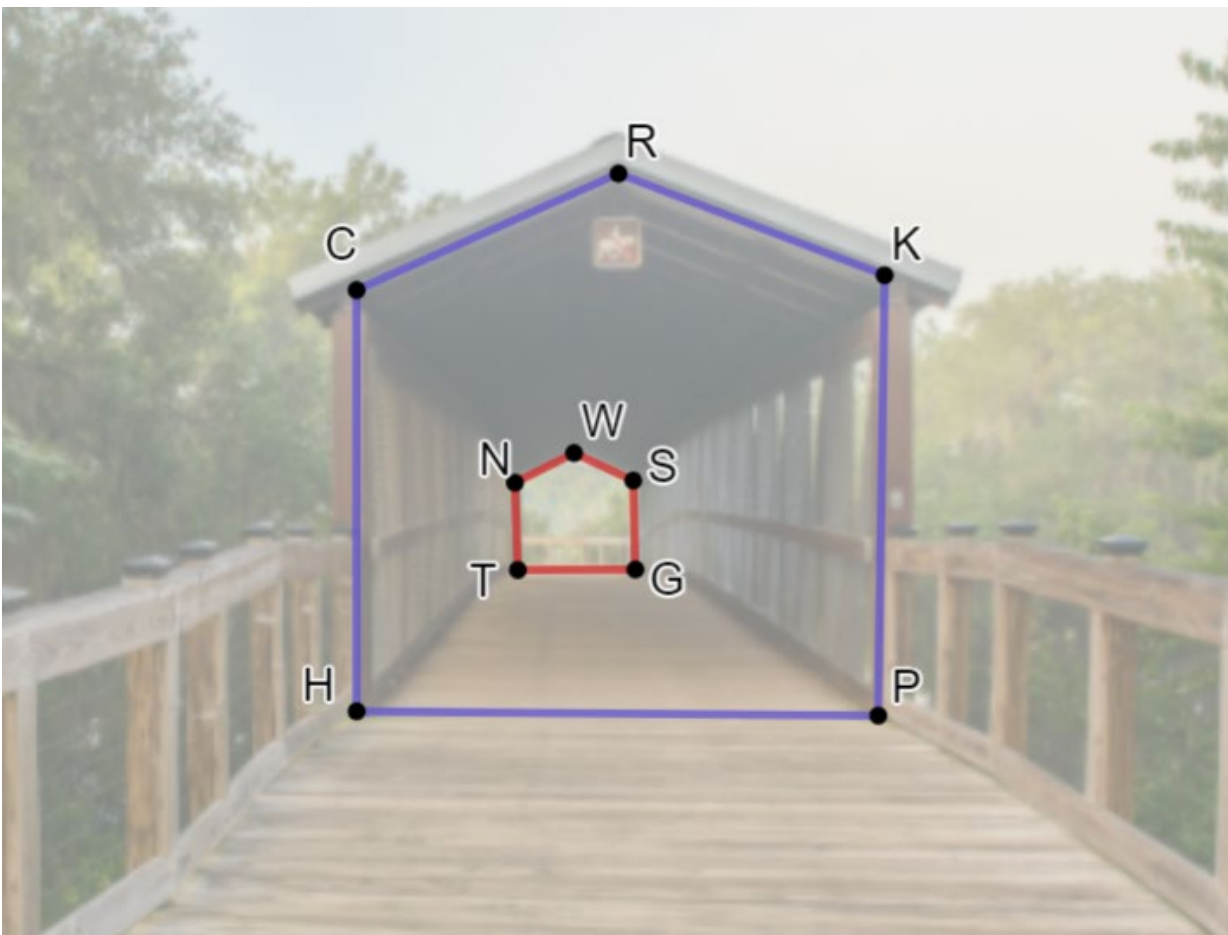
Architect George Bergstrom designed the Pentagon, which consists of five equal sides. A copy of the blueprint, *BMWRT*, represents a similar model to the actual building, *AKPSH*.



This blueprint's sides, *BMWRT*, are $\frac{1}{562}$ of the actual sides of the pentagon, *AKPSH*. If $TB = (0.5x - 5)$ meters (m) and $HA = (22x + 39)$ m, what is the perimeter of the Pentagon in meters?

Problem F

Naomi took a picture of the Lafayette Heritage Trail Park Bridge. She noticed that the perspective of the photo created two similar polygons, one at the entrance of the bridge and one at the exit of the bridge.



Naomi determines that in the photo the width of the entrance, $CRKPH$, is 4 times the size of the width of the exit, $NWSGT$. If $TG = (1.5y - 5)$ feet (ft) and $HP = (3y - 5)$ ft, what is the length of $\overline{HP}$?